Writing with AI - A Markdown + BibTeX + Pandoc Template

Uwe van Heesch (TH Köln)
Kiro (AWS)

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1 Introduction

This is a template for writing documents in Markdown, managing references in a BibTeX file, and generating a PDF (or LaTeX) with Pandoc. Edit this file (`document/paper.md`) and your bibliography (`document/references.bib`), then run the build script. See the top-level `README.md` for full instructions on macOS and Windows.

You cite sources with the Pandoc citation syntax. A single citation looks like this [1]. You can combine several in one place [2, 3]. As van Heesch and colleagues showed [1], templates save time.

2 How This Works

The source of truth is this Markdown file plus the BibTeX file. Pandoc merges them, resolves every `[@key]` into a numbered citation, and appends a formatted reference list wherever the `# References` heading appears. Change the text, rebuild, and the citations and bibliography update automatically.

2.1 Headings and Structure

Use `#` for top-level headings, `##` for subsections, and so on. Keep one blank line between paragraphs.

2.2 Lists

Bullet list:

- First point
- Second point, with a reference [3]
- Third point

Numbered list:

1. Step one
2. Step two
3. Step three

2.3 Emphasis

Use *italics* with single asterisks and **bold** with double asterisks.

2.4 Images

Embed an image with `![Caption](path)`. Keep image files in the `document/images/` folder and reference them relative to `paper.md`. Pandoc places the image and renders the caption below it:



Figure 1: An example image showing how figures are embedded and captioned.

Replace `images/example.jpg` with your own figure and update the caption.

2.5 Math

Pandoc renders LaTeX math. Write inline math between single dollar signs, e.g. the mass–energy relation $E = mc^2$, and display math between double dollar signs. For example, the arithmetic mean

of n values is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

You can use the full range of LaTeX symbols, such as Greek letters (α, β, σ), fractions, sums, and integrals like $\int_0^1 x^2 \, dx = \frac{1}{3}$.

3 Conclusion

Replace this content with your own. Keep the YAML block at the very top (it sets title, authors, margins, and font size) and keep the `# References` heading at the very bottom.

References

[1] U. van Heesch. (2026). Dear researchers: The craft that i have practiced for the last twenty years no longer exists: What happens when AI agents become operational actors in software engineering? *Journal of Systems and Software*, 242, 113000. https://doi.org/10.1016/j.jss.2026.113000.

[2] U. van Heesch, O. Zimmermann, and C. Kohls. (2026). What will remain human in software architecture? A focus group report. *Proceedings of the 31st european conference on pattern languages of programs, people, and practices (EuroPLoP 2026)*. https://doi.org/10.48550/arXiv.2609.30334.

[3] T. Theunissen, U. van Heesch, and P. Avgeriou. (2022). A mapping study on documentation in continuous software development. *Information and Software Technology*, 142, 106733. https://doi.org/10.1016/j.infsof.2021.106733.